Worldwide Public Cloud Services

Descriptions	2018	2019	2020	2021	2022
Cloud Business Process Services (BPaaS - business process as a service)	45.8	49.3	53.1	57.0	61.1
Cloud Application Infrastructure Services (PaaS - platform as a service)	15.6	19.0	23.0	27.5	31.8
Cloud Application Services (SaaS - software as a service)	80.0	94.8	110.5	126.7	143.7
Cloud Management and Security Services	10.5	12.2	14.1	16.0	17.9
Cloud System Infrastructure Services (IaaS - infrastructure as a service)	30.5	38.9	49.1	61.9	76.6
Total Market Revenue Forecast (US$ billions)	**182.4**	**214.3**	**249.8**	**289.1**	**331.2**

Source: Gartner (April 2019)

The new research report by Grand View Research, Inc mentions that, with a CAGR of 37.4 per cent during the speculated period, the market volume of edge computing is expected to reach 43.4 billion dollars by 2027 globally, where 5G and IoT would play an extremely decisive role. Various apps that will use 5G will completely change the working pattern of traffic demand and would uplift more sophisticated technology for telecom providers.

MarketsandMarkets, the platform offering market research reports, claims that by 2022 overall edge computing market would be valued around 6.72 billion dollars from a speculated 1.47 billion dollars in 2017, with a CAGR of 35.4 per cent. The best way for companies to make their top-notch IoT devices running swiftly is to amalgamate edge computing and data-gathering potential with the cloud, storage space and processing power. Apart from that, companies can also safeguard their IoT devices running efficiently against losing any imperative analytical data value.

Need of edge computing (EC) and cloud computing (CC)

On the road to cloud computing, data centres will have to modernise and extend their networks to be able to provide resources more flexibly. This changes the way networks are designed, built and monitored. As space is often very limited, ultra-high density is a 'must' for providing all required connections. The most essential benefit of edge computing is its ability to enhance network performance by reducing latency. Analysing data closer to the edge provides huge benefits in terms of cost and efficiency.

Benefits and potential of EC and CC

The US-based global tech firm Cisco predicts that by 2021, around 94 per cent of workloads would be processed and carried out by cloud data centres. It is well-known that when it comes to investment in cloud computing, the USA tops every country and by 2023, Reportlinker states, the entire cloud computing market would cross 623.3 billion dollars.

In this context, let's find out what the experts say.

Gaurav Burman. Any business using IoT devices can benefit immensely from EC and CC. Because of these new-fangled technologies, retailers are able to depend on 24/7 online presence to offer enhanced customer service, factory floor operators stay connected to plant systems, even the operation efficacies of manufacturers are enhancing, healthcare professionals have reliable access to possibly life-saving health-related data and smart buildings are redefining the way we work, live and play. From IT, financial markets and e-commerce companies to utility providers, smart cities, and governments, EC and CC are impacting nearly every business.

Anjani Kommisetti. Businesses that are customer experience-centric, or are utilising IoT, investing in 5G in their operations will benefit them from the technologies of edge and cloud computing. For instance, the automation industries have popular use cases including large engineering simulations, data and dealers' networks, equipment and data analytics, etc. In the retail sector, different information is gathered through edge computing such as customer purchase behaviour. These technologies have also greatly benefitted the healthcare sector by improving nursing efficiency through tracking of mobile medical devices, wearable fitness devices, and optimisation of medical equipment. Sectors like agriculture, BFSI, and logistics have also implemented these technologies, and businesses are realising its potential.

Adrian Johnson. Edge computing is benefitting the sector, where IoT devices are used specifically for profit purposes and utilised across-the-board in every related industry. For example: artificial intelligence (AI), smart cars, manufacturing plants, smart cities, and other 'smart' objects, and healthcare IT infrastructure are getting immense benefit from edge computing as it helps to prevent application performance latency issues. Sectors like education, law firms, emergency services, travel management, and entertainment management

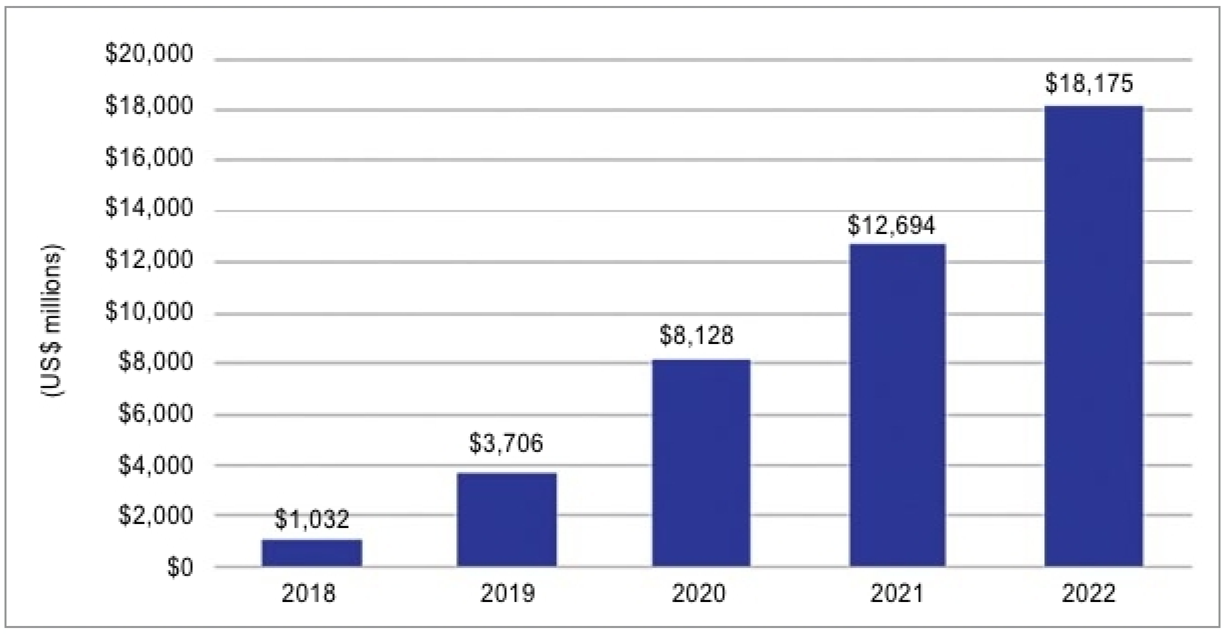

Fig. 1: Fog computing revenue, 2018-2022 (Source: 451 Research & OpenFog Consortium)